

Group Theory and Its Applications

Yi Lu

Draft of September 22, 2026

Work in progress. These notes are written up as the course runs; several chapters are still an outline only, and nothing here has been through a careful revision. Corrections are welcome.

ACKNOWLEDGEMENT. These notes draw in part on a $\LaTeX$ transcription that Kong Shuo (孔硕), then a student on this course, made from the 2025 lectures and from my handwritten notes. His careful and unprompted work is used throughout the earlier chapters and is gratefully acknowledged.

Contents

Principal texts	vii
-----------------	-----

I. Groups	1
------------------	----------

1. Introduction	3
------------------------	----------

1.1. Three themes of twentieth-century physics	3
1.2. What symmetry is, and how to measure it	3
1.3. Discrete symmetry	4
1.3.1. Mirror symmetry in a molecule	4
1.3.2. Particle exchange	5
1.3.3. Crystals	5
1.4. Continuous symmetry and Noether's theorem	6

2. Groups: basic definitions and examples	9
--	----------

2.1. Definition and first examples	9
2.2. Subgroups, order, and cyclic groups	10
2.3. Equivalence relations and residue classes	11
2.4. Direct products and Abelian groups	12
2.5. Matrix groups	12
2.6. Generators and presentations	16
2.7. Application: the Pauli group and stabilizer codes	18

3. Homomorphism and isomorphism	23
--	-----------

3.1. Homomorphism and isomorphism	23
3.2. Kernel and image	25
3.3. Examples	26
3.4. $SU(2)$ and $SO(3)$	27
3.5. $GL(V)$ and $GL(n, \kappa)$	30

4. Group actions	33
-------------------------	-----------

4.1. Actions, G -sets, and orbits	33
4.2. Cosets, Lagrange's theorem, and orbit-stabilizer	36
4.3. Equivariant maps	40
4.4. Conjugacy, centralizers, and normalizers	41
4.5. The class equation	44
4.6. Finite rotation groups	45
4.7. Induced action on function spaces	47

5. The symmetric group	49
-------------------------------	-----------

5.1. Permutations, cycles, and cycle decomposition	49
5.2. Cayley's theorem	49

5.3.	Transpositions, parity, and the alternating group	49
5.4.	Conjugacy classes of S_n and Young diagrams	49
6.	Normal subgroups, quotients, and extensions	51
6.1.	Normal subgroups	51
6.2.	Quotient groups	51
6.3.	The isomorphism theorems	51
6.4.	Exact sequences and semidirect products	51
6.5.	Central extensions and the finite Heisenberg group	51
II.	Representation theory	53
7.	Representations	55
7.1.	Motivation: why representations	55
7.2.	Basic definitions	55
7.3.	Examples of representations	55
7.4.	Operations on representations	55
8.	Unitarity and reducibility	57
8.1.	Unitary representations and the Haar measure	57
8.2.	The regular representation	57
8.3.	Reducible and irreducible representations	57
9.	Schur's lemmas and orthogonality	65
9.1.	Schur's lemmas	65
9.1.1.	Schur's lemmas I and II	65
9.2.	Orthogonality relations and the Peter–Weyl theorem	72
9.2.1.	Motivation and importance	72
9.2.2.	Preliminaries: $L^2(G)$ as a $G \times G$ representation	73
9.2.3.	The Peter–Weyl theorem	74
III.	Applications	87
10.	Crystals, spin, and time reversal	89
10.1.	Crystallographic point groups and character tables	89
10.2.	Double groups	89
10.3.	Antiunitary symmetry and Kramers degeneracy	89
10.4.	Translations, Bloch's theorem, and the Brillouin zone	89
11.	Crystal fields and molecules	91
11.1.	Character tables of the common point groups	91
11.2.	Molecular vibrations, infrared and Raman activity	91
11.3.	Splitting of atomic orbitals in a crystal field	91
11.4.	The Jahn–Teller effect	91
11.5.	Multiplets in weak and strong fields	91
11.6.	Hybridization and molecular orbitals	91

12. Space groups and band structure	93
12.1. Lattices, translations, and the space group	93
12.2. The little group and the star of k	93
12.3. Representations of space groups by induction	93
12.4. Compatibility relations and band degeneracies	93
12.5. Nonsymmorphic groups and band sticking	93
12.6. Band representations	93
13. Magnetic order and band topology	95
13.1. Magnetic point groups and corepresentations	95
13.2. Spin groups	95
13.3. Altermagnets and symmetry-enforced spin splitting	95
13.4. Inversion eigenvalues and the SSH chain	95
13.5. The Fu–Kane parity criterion	95
14. Selection rules and superconductivity	97
14.1. Tensor operators, Wigner–Eckart, and dipole selection rules . .	97
14.2. Polarisation dependence of X-ray absorption in the cuprates . .	97
14.3. Landau theory and symmetry breaking	97
14.4. Symmetry of the superconducting order parameter	97

Principal texts

- [Moore] G. W. Moore. *Abstract Group Theory. Physics 618: Applied Group Theory*. Lecture notes, Chapter 1; later chapters dated 2018–2019. 2023. URL: <https://www.physics.rutgers.edu/~gmoore/618Spring2023/GTLect1-AbstractGroupTheory-2023.pdf> (visited on 09/08/2026).
- [Zee] A. Zee. *Group Theory in a Nutshell for Physicists*. Princeton: Princeton University Press, 2016. elementary; a good entry point.
- [Dres] M. S. Dresselhaus, G. Dresselhaus, and A. Jorio. *Group Theory: Application to the Physics of Condensed Matter*. Berlin: Springer, 2008. many worked physical applications.
- [Tao] 陶瑞宝. 物理学中的群论. 高等教育出版社, 2011. a standard Chinese textbook.
- [Ma] 马中骐. 物理学中的群论. 科学出版社, 2006. a standard Chinese textbook.
- [Chen] 陈金全. 群表示论的新途径. 上海: 上海科学技术出版社, 1984. Nanjing University.